THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET A

Chapters 3 & 4: **A Story of Numbers, Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The Roman numeral CXLIX stands for: 1

(a) 150 **(b)** 149 **(c)** 159 **(d)** 139

2. 38 in Roman numerals is: 1

(a) XXXVIII **(b)** XLVIII **(c)** XXXVII **(d)** XXXIX

3. The decimal number 13 written in base 2 is: 1

(a) 1100 **(b)** 1101 **(c)** 1111 **(d)** 1110

4. The decimal number 26 written in base 3 is: 1

(a) 221 **(b)** 1000 **(c)** 1001 **(d)** 222

5. The value of 1101 in base 2, in the decimal system, is: 1

(a) 15 **(b)** 14 **(c)** 12 **(d)** 13

6. Three angles of a quadrilateral are 70°, 120° and 75°. The fourth angle is: 1

(a) 85° **(b)** 115° **(c)** 105° **(d)** 95°

7. One angle of a parallelogram is 65°. Its adjacent angle is: 1

(a) 25° **(b)** 65° **(c)** 230° **(d)** 115°

8. The diagonals of a rhombus are 6 cm and 8 cm. Its side is: 1

(a) 7 cm **(b)** 5 cm **(c)** 6 cm **(d)** 4 cm

9. The diagonal of a rectangle with sides 6 cm and 8 cm is: 1

(a) 9 cm **(b)** 14 cm **(c)** 10 cm **(d)** 11 cm

10. A quadrilateral with only one pair of parallel sides is a: 1

(a) trapezium **(b)** rhombus **(c)** parallelogram **(d)** kite

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Who named 'Hindu numerals' and who spread them in Europe? 2

12. What is a tally mark system? 2

13. Can a quadrilateral have all four angles acute? Why? 2

14. What is the sum of adjacent angles of a parallelogram? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Convert 11011_2 to decimal. 3

16. A kite has diagonals 12 cm and 16 cm. Find its area. 3

17. Convert 2201_3 to decimal. 3

OR

Convert 1234_5 to decimal. 3

18. Show that diagonals of a rectangle are equal using Pythagoras. 3

OR

A trapezium has parallel sides 10 cm and 6 cm and height 4 cm. Find its area. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Compare the Maya and Mesopotamian number systems. 5
(ii) State one similarity.

OR

(i) What is the Bakhshali manuscript? 5
(ii) What does it tell us about zero?

20. (i) In trapezium ABCD, AB ∥ CD and $\angle A = 70^\circ$, $\angle B = 80^\circ$. Find $\angle C$ and $\angle D$. 5
(ii) Why?

OR

(i) Show that the angle bisectors of a parallelogram form a rectangle (state the idea). 5
(ii) Give the angle at each corner of this rectangle.

END OF QUESTION PAPER — Total: 40 Marks

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